

YUNYANG LIU

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EDUCATION

University of California, Santa Cruz

Robotics Engineering & Electrical Engineering

Expected Jun 2026 · GPA: 3.6 / 4.0

TECHNICAL SKILLS

Programming Languages: C, C++, Python, MATLAB

Microncontroller & FPGA: PIC32MX, STM32F411RE (Nucleo), ESP32, BASYS3 FPGA

Embedded Interfaces: UART, SPI, I²C, PWM, ADC, GPIO, Interrupts

Software & Tools: Solidworks, Onshape, MPLAB X, PlatformIO, VS Code, Vivado, MATLAB, Git/GitHub, Linux CLI

Fabrication & Equipment: FDM 3D Printing (Bambu Lab P1S, P1P, H2S), PCB Soldering, xTool P2S Laser Cutter, MYT10 Automated Dispenser

EXPERIENCE

Assistive 7-DoF Robotic Manipulator for Remote Operation

Sep 2025 – Jun 2026

- ▶ Led full mechanical design (Onshape), FDM prototyping (Bambu Lab P1S), and assembly of a 7-DoF assistive robotic arm. Iterated 5+ times to refine joint tolerances and structural geometry, achieving a reliable full-arm assembly with 1.33 kg payload capacity.
- ▶ Calibrated servos performance (neutral alignment, response consistency) to improve positioning accuracy and ensure coordinated motion across 7 joints.
- ▶ Served as Mechanical Lead in a 4-person cross-functional team, aligning mechanical design decisions with electrical and software subsystems.
- ▶ Developed and executed system-level tests to evaluate motion accuracy, load capacity, and repeatability; identified tolerance stack-up issues and drove iterative design improvements.

4-DoF Sensor-Driven Robotic Manipulator (IMU + Flex Sensors)

Jan – Mar 2026

- ▶ Architected a hand gesture-controlled 4-DoF robotic arm using STM32F411RE, integrating IMU, flex, and FSR force sensors for PWM-driven actuation; achieved reliable pick-and-place in <30s (2× faster than requirement) with robust real-time sensor-to-actuation control
- ▶ Designed and fabricated mechanical components via parametric CAD (Onshape) and FDM 3D printing; iterated on designs to resolve assembly, servo clearance, and alignment issues, improving structural reliability
- ▶ Developed a power distribution system with dual buck converters and per-servo decoupling, eliminating voltage drops under >6A stall current for stable multi-servo operation
- ▶ Executed system-level tests to evaluate gesture tracking accuracy, response latency, and control stability; diagnosed sensor noise and calibration issues, and improved system robustness through iterative tuning

Sensor-Driven Embedded System Design

Jan – Mar 2026

- ▶ Developed embedded firmware in C on STM32F411RE, interfacing multi-sensor systems (IMU, encoder, ultrasonic, flex) via ADC, I2C, PWM, and interrupt-driven architecture to enable real-time sensor-driven control.
- ▶ Designed signal processing pipelines (IIR filtering, regression) to suppress sensor noise, improving stability of sensor-to-actuator mapping under dynamic conditions.
- ▶ Built interrupt-driven state machines for encoder tracking and ultrasonic sensing, boosting measurement accuracy and achieving 3.75° rotational resolution.
- ▶ Calibrated IMU sensors using bias/scale correction and ellipsoid fitting, reducing drift and significantly improving orientation accuracy and system stability.

Autonomous Event-Driven Card Dispensing System (PWM Motor Control & Sensor Feedback)

Apr – Jun 2025

- ▶ Architected a hierarchical state machine (HSM) in C (ES Framework) to coordinate servo scanning, player detection, and multi-player card dispensing, enabling robust real-time control.
- ▶ Integrated ultrasonic sensing, servo positioning, and DC motor control (L298 H-bridge) into a cohesive embedded system on a custom-soldered perfboard, ensuring reliable cross-component synchronization.
- ▶ Diagnosed and resolved multi-card dispensing failures via mechanical and control optimizations (friction guides, reverse motor logic), achieving consistent single-card output and improving system reliability.
- ▶ Designed and fabricated a custom chassis using parametric CAD (Onshape), combining FDM 3D printing and laser cutting with kerf compensation to ensure precise assembly and repeatable alignment.

Two-Wheeled Autonomous Mobile Robot

Apr – Jun 2025

- ▶ Built a keyboard-driven control interface for manual override and debugging, enabling real-time command input and stable, responsive robot control.
- ▶ Developed reactive autonomy using HSM/FSM in C (ES Framework) on PIC32, integrating bump sensors and IR phototransistors to achieve light-seeking behavior, collision-triggered direction correction, and low-light automatic stop.

Full-Duplex UART Communication Stack with Packet Protocol Jan – Mar 2025

- ▶ Implemented an interrupt-driven UART driver on PIC32 with dual circular FIFOs and a custom packet protocol (header/length/ID/payload/tail + checksum), enabling reliable full-duplex communication with validated data integrity.
- ▶ Designed an RX state machine for packet parsing and checksum verification, filtering invalid data and improving communication robustness.
- ▶ Interfaced an AS5047D magnetic encoder via SPI, decoding 14-bit angle data using a two-transaction protocol and configuring clock phase/polarity, achieving stable and accurate angle readings.

Real Estate Development Intern | Presidio Bay Ventures, San Francisco

Jun – Aug 2023

- ▶ Conducted 10+ cross-functional stakeholder interviews to assess feasibility, regulatory constraints, and financial considerations for mixed-use development projects
- ▶ Developed and presented a 45-slide mixed-use development proposal by synthesizing site analysis and stakeholder insights, supporting executive-level decision-making